

Note 2.3 Functions

Introduction

In this section, we will understand what is function and its properties.

- **Definition of Function**

Let A and B be nonempty sets. A function f from A to B is an assignment of exactly one element of B to each element of A . We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A . If f is a function from A to B , we write $f : A \rightarrow B$.

Sometimes functions are called mappings and transformations.

- **Definition of Important Concepts in Functions**

If f is a function from A to B , we say that A is the **domain** of f and B is the **codomain** of f . If $f(a) = b$, we say that b is the **image** of a and a is a **preimage** of b . The **range**, or image, of f is the set of all images of elements of A . Also, if f is a function from A to B , we say that f **maps** A to B .

And we can know the range is the **subset** of codomain.

Two functions are equal when they have the same domain, have the same codomain, and map each element of their common domain to the same element in their common codomain.

- **Definition of Real-Valued and Integer-Valued Functions and Its Computation**

A function is called real-valued if its **codomain** is the set of real numbers, and it is called integer-valued if its codomain is the set of integers.

Let f_1 and f_2 be functions from A to $\mathbf{R}$ or $\mathbf{Z}$, then $f_1 + f_2, f_1 f_2$ are also functions from A to $\mathbf{R}$ or $\mathbf{Z}$ defined for all $x \in A$ by

$$(f_1 + f_2)(x) = f_1(x) + f_2(x), (f_1 f_2)(x) = f_1(x) f_2(x)$$

- **Definition of Image of Sets**

Let f be a function from A to B and let S be a subset of A . The image of S under the function f is the subset of B that consists of the images of the elements of S . We denote the image of S by $f(S)$, so

$$f(S) = \{t | \exists s \in S (t = f(s))\}$$

or $\{f(s) | s \in S\}$.

One-to-One and Onto Functions

- **Definition of One-to-One Function**

A function f is said to be one-to-one, or an **injection**, if and only if $f(a) = f(b)$ implies that

$a = b$ for all a and b in the domain of f . A function is said to be **injective** if it is one-to-one.

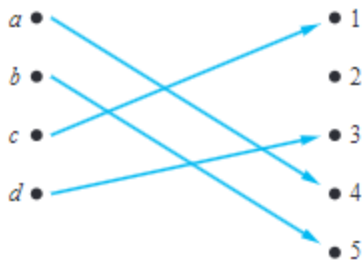


FIGURE 3 A One-to-One Function.

That is,

$$\forall a \forall b (f(a) = f(b) \rightarrow a = b) \text{ or } \forall a \forall b (a \neq b \rightarrow f(a) \neq f(b))$$

- **Definition of Increasing and Decreasing Functions**

A function f whose domain and codomain are subsets of the set of real numbers is called increasing if $f(x) \leq f(y)$, and strictly increasing if $f(x) < f(y)$, whenever $x < y$ and x and y are in the domain of f . That is, increasing if $\forall x \forall y (x < y \rightarrow f(x) \leq f(y))$ and strictly increasing if $\forall x \forall y (x < y \rightarrow f(x) < f(y))$.

Similarly, f is called decreasing if $f(x) \geq f(y)$, and strictly decreasing if $f(x) > f(y)$, whenever $x < y$ and x and y are in the domain of f . (The word strictly in this definition indicates a strict inequality.) That is, increasing if $\forall x \forall y (x < y \rightarrow f(x) \geq f(y))$ and strictly increasing if $\forall x \forall y (x < y \rightarrow f(x) > f(y))$.

- **Definition of Onto Function**

A function f from A to B is called onto, or **surjection**, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$. And a function f is called **surjective** if it is onto. And we can know that the codomain and the range are the same set.

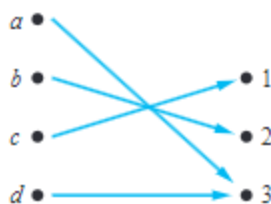


FIGURE 4 An Onto Function.

That is,

$$\forall y \exists x (f(x) = y)$$

- **Definition of One-to-One Correspondence**

The function f is a one-to-one correspondence, or a **bijection**, if it is both one-to-one and onto. We also say that such a function is **bijective**.

Inverse Functions and Compositions of Functions

- **Definition of Inverse Functions**

Let f be a one-to-one correspondence from the set A to the set B . The inverse function of f is the function that assigns to an element b belonging to B the unique element a in A such that $f(a) = b$. The inverse function of f is denoted by f^{-1} . Hence, $f^{-1}(b) = a$ when $f(a) = b$. Sometimes we can restrict the domain or the codomain of a function, or both, to obtain an invertible function.

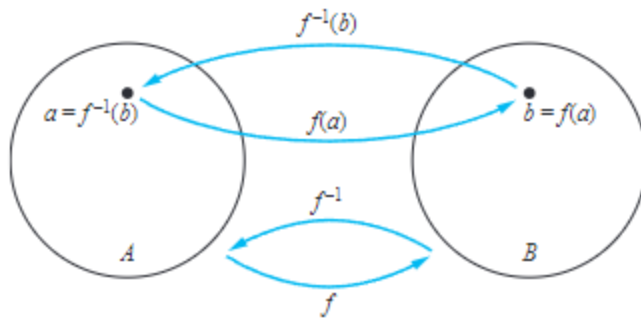


FIGURE 6 The Function f^{-1} Is the Inverse of Function f .

- **Definition of Composition of functions**

Let g be a function from the set A to the set B and let f be a function from the set B to the set C . The composition of the functions f and g , denoted for all $a \in A$ by $f \circ g$, is defined by

$$(f \circ g)(a) = f(g(a))$$

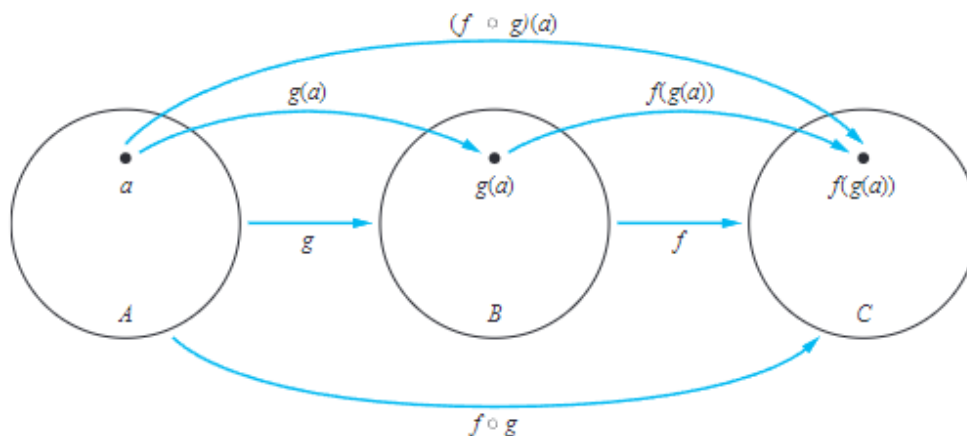


FIGURE 7 The Composition of the Functions f and g .

The Graphs of Functions

- **Definition of Graph**

Let f be the function from the set A to the set B , the graph of the function f is the set of

ordered pairs $\{(a, b) | a \in A \text{ and } f(a) = b\}$.

Some Important Functions

- **Floor Function and Ceiling Function**

The floor function assigns to the real number x the largest integer that is less than or equal to x . The value of the floor function at x is denoted by $\lfloor x \rfloor$. It is also called the greatest integer function, denoted by $[x]$.

The ceiling function assigns to the real number x the smallest integer that is greater than or equal to x . The value of the ceiling function at x is denoted by $\lceil x \rceil$.

- **Factorial Function**

$f : \mathbf{N} \rightarrow \mathbf{Z}^+, f(n) = n!$.

Partial Functions

- **Definition of Partial Functions**

A partial function f from a set A to a set B is an assignment to each element a in a subset of A , called the domain of definition of f , of a unique element b in B . The sets A and B are called the domain and codomain of f , respectively. We say that f is undefined for elements in A that are not in the domain of definition of f . When the domain of definition of f equals A , we say that f is a **total function**.